

Mohamed Ali Bouzir

Software Engineer

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Innovative Software Engineer with 3+ years of experience building high-performance web applications using React.js/Next.js and Express.js. Proven expertise in API development, system architecture optimization, and delivering scalable solutions that handle 1000+ concurrent users. Strong background in performance optimization, with experience in modern frameworks including Next, React and Nest.js.

Experience

MAYRA Entertainment

Jan 2026 – Present

Software Architect & Lead Engineer - Contract

Sousse, Tunisia

- Designed and owned the end-to-end system architecture of a large-scale real-time financial platform, covering module boundaries, data flow contracts, and future microservices migration path.
- Architected a multi-tenant data isolation layer using a schema-per-tenant PostgreSQL strategy, enabling seamless future migration to a fully isolated microservices topology without application-level rewrites.
- Architected a zero-trust authentication system using opaque session tokens stored in Redis, HttpOnly/Secure/Strict cookies, and server-side session invalidation, eliminating XSS token-theft attack surface entirely.
- Designed a 4-layer wallet mutation security model combining idempotency guards, distributed Redis locks, optimistic concurrency versioning, and atomic SQL constraints to prevent double-spend and race conditions under concurrent load.
- Built an asynchronous audit pipeline using BullMQ and Redis, decoupling all logging and observability concerns from the request lifecycle without adding latency to user-facing responses.
- Led backend and frontend engineering teams across the full stack (NestJS, Next.js, PostgreSQL, Redis, Prisma), enforcing strict module encapsulation rules and cross-module communication contracts via EventEmitter and BullMQ.
- Established platform-wide architectural governance: no direct cross-module Prisma access, DTO-only boundaries, and event-driven side effects, reducing coupling and enabling independent team delivery.
- Architected a real-time engine with atomic Lua scripts in Redis ensuring transactions are frozen at payment time, combined with exposure monitoring enforcement to control financial risk.
- Produced formal technical documentation including technology selection reports, integration specification letters, and architectural decision records to support regulatory and banking compliance workflows.
- Integrated third-party payment infrastructure, designing the deposit flow from frontend initiation through backend confirmation to atomic wallet with full traceability.
- Implemented a comprehensive E2E and unit test suite covering auth lifecycle, session management, and concurrent wallet operations to guarantee correctness under adversarial conditions.

Vivia-M

Feb 2024 – Present

Software Engineer - Full-time

Sousse, Tunisia

- Optimized application performance by 24% through advanced profiling and optimization techniques
- Re-designed app architecture to ensure scalability and maintainability reducing development costs by 10%.
- Implemented new features using React and Redux, resulting in a 8% increase in user engagement within the first month.
- Reviewed, optimized, and fixed critical cron job workflows responsible for invoice generation and recurring payment processing, ensuring reliable scheduled operations, payment consistency, and improved stability of automated billing systems.
- Enhanced database performance by optimizing MySQL queries and implementing parallel execution, reducing API response times by 15%.
- Built high-performance APIs that handle storage efficiently and are resistant to spam and malicious traffic.
- Integrated Brevo and configured dynamic templates for sending personalized emails.
- Developed and maintained all dashboard forms using React Hook Form, ensuring type-safe inputs, real-time validation, and seamless API integration, which improved user experience and reduced form submission errors by 30%.
- Implemented comprehensive SEO strategy using React Helmet, collaborating with marketing team.
- Worked with Agile methodology (Scrum).

Billcom Consulting

Jan 2023 – Aug 2023

Software Engineer - Full-time

Sousse, Tunisia

- Designed and implemented a high-performance data migration tool from scratch, designed to handle complex migration workflows.
- Developed algorithms to address performance-related challenges, ensuring optimal data migration speed and resource usage.
- Generated complex SQL queries based on available resources to extract data efficiently during migration.
- Implemented deep level services with loosely coupled components to enable parallel data migration, improving performance and scalability.
- Designed a versatile solution with a command-line interface, a web interface, and a desktop application.
- Designed and developed a dashboard using React.js with the ability to initiate migrations and visualize the generated reports.
- Incorporated Progressive Web App (PWA) specifications to enable the installation feature for the application, improving user accessibility.

Digital Market

Jun 2022 – Sept 2022

Ionic/React Developer - Freelance

Monastir, Tunisia

- Architected and developed a cross-platform mobile application using **Ionic Capacitor**, ensuring a native-like experience and seamless hardware integration across iOS and Android.
- Engineered complex, type-safe data visualization dashboards using **React TypeScript**, optimizing the rendering of large datasets with advanced state management.
- Implemented secure client-side **JWT authentication** workflows and protected routing, maintaining robust security standards across web and mobile interfaces.

SkillWare

Jan 2022 – Dec 2022

Front-End Developer - Part-time

Sousse, Tunisia

- Developed user-friendly, responsive interfaces using React, TypeScript.
- Collaborated with UI/UX designers to translate Figma designs into pixel-perfect responsive interfaces.
- Utilized React best practices, including the HOC design pattern, lazy loading, suspense, memos, and callbacks to improve PSI KPIs by 17%.
- Improved app performance by minimizing unnecessary re-renders with efficient context usage and memoization in custom hooks, resulting in a 25% reduction in load times.

Code Time

Jul 2021 – Oct 2021

Software Automation QA Engineer - Internship

Tunis, Tunisia

- Developed a Selenium-based automation framework to test a platform with over 2000 pages, covering all buttons, input fields, filters, and navigation paths without manual intervention.
- Implemented automated bug reporting capturing HTTP status codes, error locations via XPath, and generating comprehensive structured reports for each test run.
- Automated detection of page load failures, broken filters, incorrect query results, status code anomalies, invalid redirections, and broken routing paths.
- Optimized script performance using multithreading to parallelize inter-test wait times, reducing total test duration from 30 hours to 19 hours.
- Further reduced execution time to 25 minutes by introducing multiprocessing to handle multiple pages simultaneously, achieving a 98% reduction in total testing time.
- Deployed a cron job scheduling automated full-suite test runs twice daily at 07:00 and 18:00, ensuring continuous regression coverage.

Technical Skills

Programming Languages: JavaScript, TypeScript, Python

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, Material UI, TailwindCSS

Web Technologies: HTML5, CSS3, RESTful APIs, GraphQL

Databases: Oracle, MySQL, MongoDB, PostgreSQL

Development Tools: Docker, Postman, VS Code

Version Control: Git, GitLab, GitHub

Project Management: Jira, Trello, Confluence

Methodologies: Agile Scrum

Others: Object-Oriented Programming (OOP), Design Patterns

Education

Software Engineering Degree in Computer Science (GPA: 4.00 / 4.00)

The Higher Institute of Applied Science and Technologies

Sept 2020 - May 2023

Sousse, Tunisia

Bachelor Degree in Computer Science (GPA: 3.70 / 4.00)

The Higher Institute of Applied Science and Technologies

Sept 2018 - May 2020

Sousse, Tunisia

Projects

National Digital Transformation Platform (Confidential State Project – NDA Restricted)

Led the architecture and full-stack engineering of a confidential public-sector platform, supervising backend and frontend teams and defining the system's technical direction. Implemented a domain-driven modular monolith architecture optimized for scalability and future migration toward microservices, with emphasis on clean boundaries, maintainability, secure integrations, and enterprise-grade operational reliability.

- **Keywords:** *React, Next.js, Typescript, Orval, Tanstack, Zustand, Tailwind CSS, shadcn.ui, Framer-motion, Nest.js, Prisma, PostgreSQL, Redis, BullMQ, Docker, CI-CD*

Effia BE

High-traffic parking management system serving 1000+ active users with real-time space reservations, online payments, and availability tracking. Built comprehensive admin dashboard for staff management of parking spaces, user roles, pricing, and billing systems.

- **Keywords:** *React.js, Typescript, Material-UI, Redux, socket.io, Express.js, Node.js, Iodash, Javascript, MySQL, Git, Gitlab, Agile*

Effia Dashboard

Aladin Staff-only restricted portal for managing the Effia BE application, covering parking space lifecycle (insertion, updates, deletions), user role management, pricing configuration across logical and legal entities, billing, offers, reservations, and subscription tracking.

Jasmin Internal finance module dedicated to invoice management and full payment traceability, enabling staff to audit and track all transaction flows across the platform.

Jafar PMS integration dashboard handling hardware connectivity and gateway integration for physical access control infrastructure.

- **Keywords:** *React.js, TypeScript, Material-UI, Redux, Express.js, Node.js, Lodash, JavaScript, MySQL, Git, GitLab, Agile*

DataMigrationMorpher

Comprehensive data migration solution with web, mobile, and shell interfaces. Automated entire migration process, eliminating manual intervention while providing progress tracking and issue detection.

- **Keywords:** *Python, ETL, OLAP, OLTP, SQL, Oracle, Cx Oracle, FastAPI, WebSocket, React, TypeScript, React-hook-form, PWA, Git, MultiProcessing, MultiThreading, Signals*

Carthagobet

A web app for online sports betting and casino games, giving users a simple and fun way to place bets and enjoy various casino games.

- **Keywords:** *React, Javascript, SWR, Tailwind CSS, Daisy UI, GraphQL, Apollo, MongoDB, Git, Agile*

Keywords: *React, JavaScript, Apollo, framer-motion, i18n, SASS, React-Typed, Git*

Languages

Arabic: Native | **English:** Advanced | **French:** Advanced | **German:** Beginner